

Monitoring and predicting influenza epidemics from routinely collected severe case data

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Abstract

Influenza remains a significant burden on health systems. Public health responses should be tailored to the size and timing of any ongoing outbreak. For monitoring purposes, data on severe cases of influenza in England are reported weekly to Public Health England. These data are both readily available and have the potential to provide powerful information for inferring and predicting the features of seasonal and pandemic influenza.

We propose an epidemic model which links the underlying transmission dynamic process to data on severe influenza cases. Within a Bayesian framework, we infer the parameters of the epidemic model for each seasonal outbreak from 2012 to 2015, including the effective reproduction number, initial susceptibility, the probability of admission to Intensive Care given infection and the attack rate.

The epidemic model is also implemented in real time to assess whether early forecasting is possible. Simulation exercises show that the epidemic model is able to forecast the development of seasonal outbreaks once data from the first three months are available. This paper suggests that severe case data may be effectively exploited to estimate epidemiological characteristics and to predict the further evolution of an epidemic.

Keywords: Epidemic monitoring; Bayesian inference; Epidemic models; Influenza; Reproduction number; Severe cases.

1 Introduction

Recent annual epidemics of influenza have resulted in about 3 to 5 million cases of severe illness each season worldwide [1]. Historically, influenza has always placed a large burden on many national health systems [2], particularly as a result of severe cases in the most at risk groups [3].

Measures of different characteristics of an outbreak, whether seasonal or a newly emergent strain, are crucial for understanding and planning for the healthcare burden. For seasonal influenza, retrospective knowledge of severity is necessary for having a baseline measure against which to compare the severity of any future pandemic; and prospectively, prediction of the likely extent of transmission and thus the projected number of severe cases are required to plan for the number of hospital beds needed each season. For a new emerging strain with the potential to cause a pandemic, real-time estimation of severity and prediction of transmission is even more crucial for informing a targeted response [4].

Epidemic models are often used to estimate transmission aiming to design effective responses such as: vaccination policies [5]; school closures to mitigate transmission in a pandemic ([6–9]); reinforced use of antiviral drugs; or changes in hospital management policies.

Many type of transmission model have appeared in the literature. For example, transmission can be estimated over different scales, with some studies exploiting air traffic data to estimate worldwide transmission [10, 11], while others formulate household models [3, 12]. Another variation is the type of data used: transmission is often inferred from samples of General Practitioner (GP) consultations for influenza [13–15] or Influenza Like Illness (ILI) [5, 16]. However, GP data may be difficult to access in real-time, and are typically affected by observational noise. Other studies retrospectively analyse data on all severe cases [8, 17, 18], focussing on transmission and severity during the 2009 pandemic.

Following the 2009 pandemic, the World Health Organization (WHO) declared the beginning of a post pandemic phase [19], encouraging national public health agencies to establish hospital-based surveillance systems to monitor the epidemiology of severe influenza. In response to these guidelines, and to achieve the dual aims of understanding the baseline epidemiology of seasonal influenza for comparison with future pandemics and prediction of the impact of both seasonal and pandemic influenza on the health system, the UK developed a surveillance system to monitor

severe cases of influenza, the UK Severe Influenza Surveillance System (USISS) [20, 21]. After a pilot phase in 2010/11, USISS has run for each influenza season, providing data on laboratory-confirmed Intensive Care Unit (ICU)/High Dependence Unit (HDU) influenza cases and on laboratory-confirmed hospitalized cases.

Up to now not much attention has been paid to estimating seasonal, as well as pandemic, transmission from severe case data. Such data are more readily available through systems such as USISS than GP consultation data, for example, and can provide a census, rather than a sample, of severe, ICU-admitted, cases. We therefore investigate if data collected through USISS can inform the features that characterise seasonal and pandemic epidemics, aiming to achieve both the estimation and prediction goals.

We formulate an epidemic model that links the available USISS data to the underlying dynamics of influenza in the UK. The model parameters are inferred using data from the seasonal epidemics in 2012-2015, to obtain national estimates of transmission, as measured by R_n , the average number of new cases generated by an infectious individual in a partially immune population, and severity, as measured by the probability of ICU admission given infection.

Additionally, to assess the predictive power of the model, we perform analyses at different dates within each season. Finally, we study what would happen in the event of a pandemic, when the USISS surveillance scheme is upgraded to collect more information.

The paper is structured as follows: in Section 2 the USISS data and their method of collection are introduced. In Section 3 we propose our epidemic model. In Section 4 we report the main results obtained and in Section 5 we conclude with a discussion.

2 Data

The existing UK Severe Influenza Surveillance System (USISS) provides data on weekly severe cases of influenza according to two schemes: a mandatory scheme and a sentinel scheme. Data are collected using a web tool where hospital Trusts report all admissions during the previous week (Monday 00:00 to Sunday 23:59) on each Wednesday, by 11:00. Both the schemes are run annually from week 40 to week 20 of the following year but, in the event of a pandemic, they can be activated out of this window. These schemes are briefly summarized below.

2.1 USISS mandatory scheme

According to the USISS protocol [20], all National Health Service (NHS) Trusts have to report the weekly number of laboratory-confirmed influenza cases admitted to ICU/HDU and the number of confirmed influenza deaths in ICU/HDU. An ICU/HDU case is defined as a person who is admitted to ICU/HDU and has a laboratory-confirmed influenza-A (including H1, H3 or novel) or B infection. A confirmed fatality is defined as a person who dies while in ICU/HDU and has a laboratory-confirmed influenza A (including H1, H3 or novel) or B infection. In the event of a pandemic, the same data are collected at all levels of care, not only ICU/HDU.

Data are broken down by age group and influenza type/subtype. Total ICU/HDU admissions between 2012 and 2015 are shown in Figure 1.

Data are very different across seasons. In the 2012/13 season, mainly characterized by Influenza B and Influenza A(H3N2) outbreaks, data peak early, maintaining this plateau for several months [22]. In 2013/14, when the predominant strain was A(H1N1), the time series displays a smoother increase, a well localized peak and a successive regular decrease [23]. Lastly, in 2014/15, the number of ICU admissions peaks earlier, has a dramatic drop at the beginning of the new year, which is followed by a smaller wave. This leads to a double peak in the time series. During this seasons, Influenza A(H3N2) was the predominant virus circulating and the total number of

ICU admissions is higher, since this strain is known to lead to more severe outcome, particularly in the elderly [24].

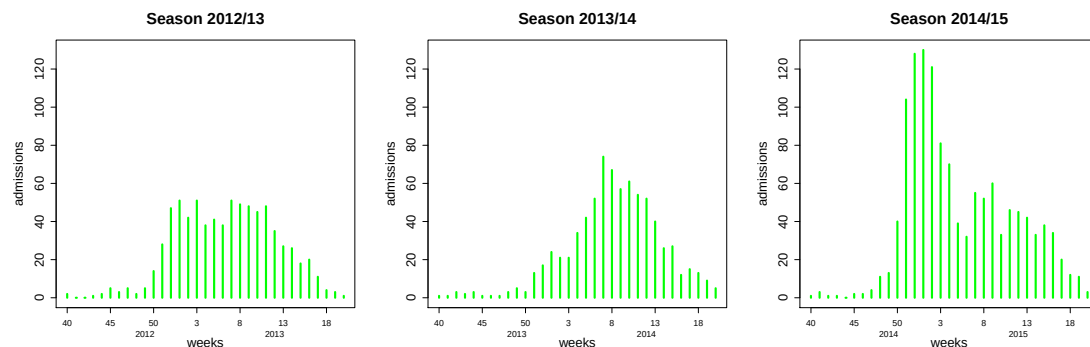


Figure 1: Aggregate weekly ICU/HDU admissions in the years analysed.

2.2 USISS Sentinel scheme

In addition to the mandatory scheme, a subgroup of NHS Trusts in England is recruited every year to participate in a sentinel scheme [21, 25]. Enrolled Trusts report the weekly number of laboratory confirmed influenza cases hospitalised at all levels of care. Data are broken down by age group and type/subtype. Where available, further epidemiological data on recorded patients are also reported. To ensure representativeness, 36 Acute Trusts are recruited by stratified random sampling. Trusts are stratified by region, size (<500 beds and ≥ 500 beds) and Trust type (Acute or Teaching). Three Trusts (one small Acute Trust, one large Acute Trust and one Teaching Trust) are randomly chosen to participate from each region in England. In London and the North West, which have a denser population, six Trusts are recruited instead of three. If a chosen Trust declines to participate, another Trust that is of the same size and type and from the same region is chosen at random and contacted. The process continues until the desired number and distribution of Trusts is reached.

Two levels of data are collected. The first is aggregate hospital-wide data: the weekly count of the number of laboratory confirmed influenza cases hospitalised and died at all levels of care is reported. The second is individual-level data on all ICU/HDU admissions (until season 2012/13) or on hospital admissions in the young (≤ 17 years old) population (from season 2013/14 onwards) (see Supplementary Information, Section SI 2, for full specification and analyses of the USISS sentinel scheme).

3 Methods

We used epidemic models to describe the transmission dynamics of influenza in the population (e.g. [26–28]). These methods are briefly outlined below, with the transmission component described in section 3.1, and the observational model in section 3.2. Computation methods are addressed in subsection 3.3.

3.1 Transmission model

Transmission models [28] divide the population into compartments, according to their health status with respect to the virus considered. The number of people in each compartment evolves over time. We assume a discrete-time deterministic SEIR-type model (see Figure 2), with time divided into intervals of length δt of a quarter of a day, i.e. every interval elapses from t to $t + \delta t$. At each t , the whole population of size N is divided into Susceptible ($S(t)$), Exposed ($E(t)$), Infectious ($I(t)$) and Removed ($R(t)$) compartments. To allow more flexibility for the time spent in E and I , we divide the E and I compartments into two sub-compartments ($E_1(t)$, $E_2(t)$ and $I_1(t)$, $I_2(t)$ respectively) [13].

Transitions between states is governed by parameters: σ is the daily rate of transition from the compartment E_1 to E_2 and from E_2 to I_1 , leading to an average incubation period $d_L = 2\frac{1}{\sigma}$ days; γ is the daily rate of transition from the compartment I_1 to I_2 and from I_2 to R leading to an average infectious period $d_I = 2\frac{1}{\gamma}$ days; and $\lambda(t)$ is the rate at which a person becomes infected, assumed to be proportional to the number of infectious people at time t and the constant transmission rate β (the rate at which new infections take place given a contact), as expressed in equation (1).

$$\lambda(t) = \beta \frac{I(t)}{N} \quad (1)$$

Here we have assumed homogeneous mixing among contacts (i.e. people are all equally likely to meet, irrespective of their age class and residence, for example) since observed severe cases are too sparse to infer a transmission matrix among age groups.

The deterministic model is described by the system in Equation (2).

$$\begin{cases} S(t + \delta t) = S(t) - \lambda(t)\delta t \cdot S(t) \\ E_1(t + \delta t) = E_1(t) - \sigma\delta t \cdot E_1(t) + \lambda(t)\delta t \cdot S(t) \\ E_2(t + \delta t) = E_2(t) - \sigma\delta t \cdot E_2(t) + \sigma\delta t \cdot E_1(t) \\ I_1(t + \delta t) = I_1(t) - \gamma\delta t \cdot I_1(t) + \sigma\delta t \cdot E_2(t) \\ I_2(t + \delta t) = I_2(t) - \gamma\delta t \cdot I_2(t) + \gamma\delta t \cdot I_1(t) \\ R(t + \delta t) = R(t) + \gamma\delta t \cdot I_2(t) \end{cases} \quad (2)$$

Since we assume a deterministic model, the whole system (i.e. values of the states $S(t)$, $E_1(t)$, $E_2(t)$, $I_1(t)$ and $I_2(t)$ for all t in $[0, T]$) is fully determined by the definition of the transition rates (β , σ and γ) and the initial state of the system ($S(0)$, $E_1(0)$, $E_2(0)$, $I_1(0)$, $I_2(0)$, $R(0)$).

Under the assumption that the epidemic is growing exponentially at its start, the model can be re-parametrized [13, 29]. Let ψ denote the Exponential Growth Rate (EGR), i.e. the rate at which the compartments E_1 , E_2 , I_1 and I_2 grow. Then under the EGR assumption at time $t = 0$, the dynamic system is determined by the EGR ψ , the proportion of the population who isre initially susceptible π , the initial infection rate λ_0 and the transition rates σ and γ . A number of quantities may be defined, therefore, in terms of these parameters, including: the transmission rate β ; the initial number of infectious people (denoted by I_0^{tot}); the basic reproduction number R_0 that is the average number of successful transmissions per infectious person in a fully susceptible population; and the effective reproduction number R_n that is the average number of successful transmissions per infectious person in partially susceptible population. The formulation of R_0 under the EGR assumption is:

$$R_0 = \psi \cdot d_I \cdot \left\{ \frac{\left(\frac{\psi d_L}{2} + 1 \right)^2}{1 - \frac{1}{\left(\frac{\psi d_I}{2} + 1 \right)^2}} \right\} \quad (3)$$

Full specification of the EGR parametrisation is reported in the Supplementary Information, Section SI 1.1.

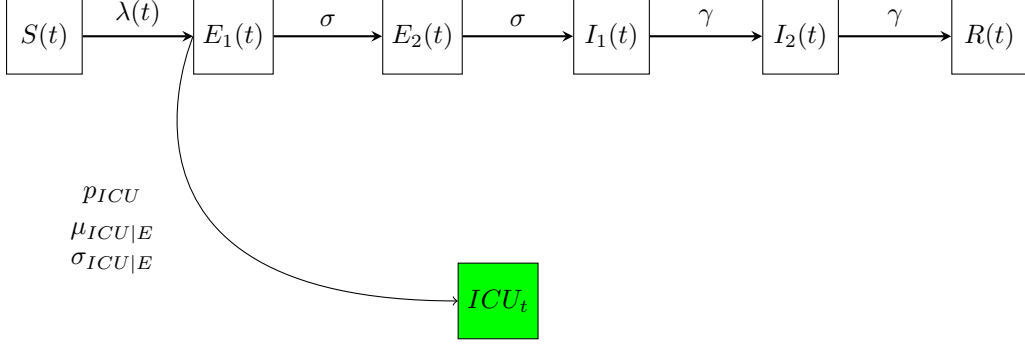


Figure 2: Compartmental model for influenza transmission dynamics and ICU/HDU admissions.

3.2 Observational model

Routinely collected ICU/HDU admissions data are linked to the transmission model through an observational model [30] which is represented in the bottom part of Figure 2.

We assume that only a proportion p_{ICU} of the people that become infected are admitted to ICU/HDU. This event happens after a waiting period that is approximated by a Gamma-distributed random variable with mean $\mu_{ICU|E}$ and variance $\sigma_{ICU|E}^2$. And therefore μ_t , the expected number of ICU/HDU admissions during interval $(t; t + \delta t]$, can be linked to the transmission model via a convolution: we express μ_t as a function of the infections that have happened in the past and the distribution of the time from infection to ICU admission [31] (see Supplementary Information, Section SI 1.2).

We assume that observations on the weekly count of ICU/HDU admissions (denoted by ICU_w , where w indexes the week) are distributed according to a Negative Binomial random variable centred on the weekly mean ICU/HDU admissions μ_w , where $\mu_w = \sum_{t \in w} \mu_t$, with dispersion parameter η [13].

$$ICU_w \sim \text{NegBin}(\mu_w, \eta) \quad (4)$$

Equation (4) allows inference of the parameter p_{ICU} , the probability that someone infected with influenza is admitted to ICU.

3.3 Estimation

The set of parameters that characterizes the transmission model is $\Theta_{\mathbf{T}} = (\pi, \lambda_0, \psi, \sigma, \gamma)$. Of these, σ and γ are assumed known from previous studies [13, 32], as these parameters can be inferred only with detailed information at the individual level. The set of parameters that determines the observational model is $\Theta_{\mathbf{O}} = (\mu_{ICU|E}, \sigma_{ICU|E}^2, p_{ICU})$, with $\mu_{ICU|E}$ and $\sigma_{ICU|E}^2$ known from previous studies [33] and/or derived from the USISS sentinel scheme (see Supplementary Information, Section SI 2). Therefore the set of unknown parameters is $\Theta = (\pi, \lambda_0, \psi, \eta, p_{ICU})$. We assume a fixed population over each season of size N_{year} , obtained from official published estimates [34].

We estimate the unknown parameters Θ within a Bayesian framework [35]. We assume each parameter distributes according to a random variable on which we have *a priori* knowledge. We combine this knowledge, formalized by defining prior distributions on each element of Θ , with the likelihood of the data to obtain posterior distributions of the parameters. These summarize both our previous knowledge and the information carried by the data. To sample from the posterior distribution of Θ , we formulate a Metropolis Hastings algorithm [36], whose specification is described in detail in the Supplementary Information, Section SI 1.3. Summaries of the posterior distributions are presented as medians and 95% credible intervals (CrI) of the samples.

4 Results

Results from analysing retrospectively the time series of each season are reported in Section 4.1. Results from “real-time” analyses performed while the epidemic is still ongoing are given in Section 4.2. Lastly, in Section 4.3, we report the results of the analysis of a simulated pandemic.

Unknown parameters definition	Parameter	Distribution
Susceptibility	π	$\sim Unif(0, 1)$
Initial infection rate	λ_0	$\sim Unif(0, 0.000241)$
Exponential growth rate	ψ	$\sim Unif(0, 0.39)$
Over-dispersion	η	$\sim Unif(1, 100)$
Probability of ICU admission given infection	p_{ICU}	$\sim Unif(0, 1)$
Parameters assumed known	Parameter	Value
Rate of becoming infectious	σ	1
Rate of recovery	γ	0.5797
Mean time from infection to ICU admission	$\mu_{ICU E}$	5.708
Variance of time from infection to ICU admission	$\sigma_{ICU E}^2$	18.24
Population of 2012/13	$N_{2012/13}$	53,679,750
Population of 2013/14	$N_{2013/14}$	54,091,200
Population of 2014/15	$N_{2014/15}$	54,551,450

Table 1: Flat prior distributions and fixed parameters for the retrospective analysis of seasonal time series.

4.1 Retrospective analysis of seasonal time series

Initially, we chose to use uniform prior distributions for all the parameters to assess whether the USISS data are informative enough to estimate all elements of Θ .

To derive the prior distributions for λ_0 and ψ we assumed an initial total number of infectious people $1 \leq I_0^{\text{tot}} \leq 10,000$ and a transmission rate $0 \leq \beta \leq 1.25$, leading to a basic reproduction number $1 \leq R_0 \leq 4$. These limits were chosen according to previous literature on earlier influenza epidemics [37]. Prior distributions and fixed parameters are reported in Table 1.

Resulting medians and 95% Credible Intervals of the posterior distributions of the parameters are reported in Table 2.

There is high correlation among some of the elements of Θ (namely π , ψ and p_{ICU}). Moreover, results show that USISS data do not contain enough information to estimate all the parameters separately. More specifically, the model used requires separate information to identify both the initial state and growth of the epidemic. However, we have only one source of information,

linked to the transmission dynamics via a waiting time process: these the data are not sufficient to identify separately to the two processes. As a consequence, the posterior distribution of the unidentifiable parameters would be identical to their prior. This problem has been referred as lack of identifiability [38]. Its diagnosis and consequences in our analysis is explore in the Supplementary Information, Section SI 3.

However, other elements of Θ and functions of the unidentifiable parameters were still estimable. One of these quantities is the effective reproduction number R_n , a function of both the susceptibility π and the EGR ψ . Posterior distributions of R_n for the three epidemics analysed

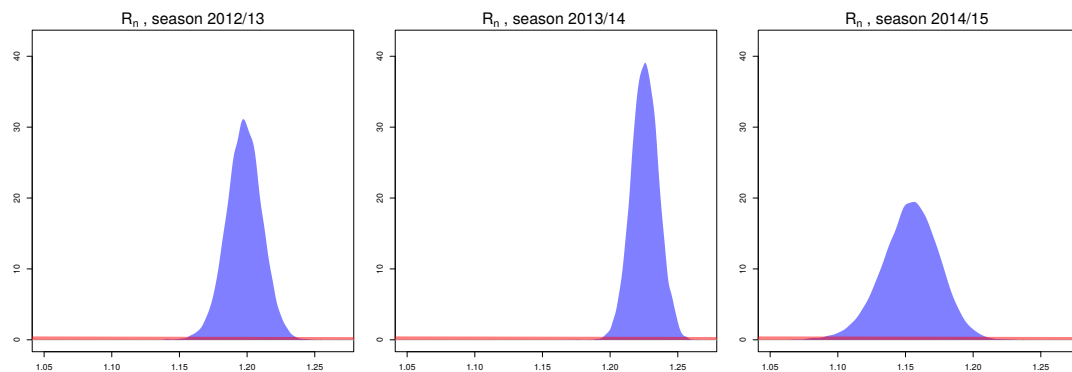


Figure 3: Prior (red) and posterior (blue) distributions of R_n across seasons .

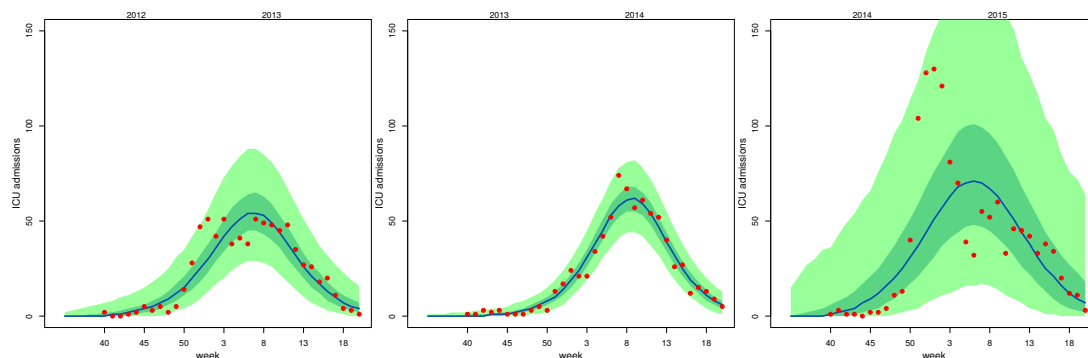


Figure 4: Median (blue), 95 % Credible Intervals (light green) and quartile (dark green) of the posterior predictive distributions and observed values (red) of the weekly ICU/HDU admissions across seasons.

are shown in Figure 3. The estimated medians (95% CrI) for R_n are: 1.198 (1.171-1.223) for 2012/13, 1.226 (1.206-1.246) for 2013/14 and 1.154 (1.110-1.193) for 2014/15. Compared to 2012/13, during the 2013/14 outbreak, the infection is more transmissible. During 2014/15, the estimate if R_n is very variable due to the double peak of the data that our model does not account for.

To evaluate the fit of the model we estimated the distributions of the weekly ICU/HDU admissions (i.e. the posterior predictive distribution) and compared them to the observed values. These plots are shown in Figure 4. Both Figures 3 and 4 highlight the large variability in the estimates obtained for season 2014/15. The model we formulated was not able to reproduce the

double peak seen in these data. To compensate this constraint, estimates of the over-dispersion parameter η are notably larger than the ones obtained from the data on the 2012/13 and 2013/14 seasons (see results in Table 2). On the other hand, during the earlier epidemics (2012/13 and 2013/14), the data are well reproduced and the posterior distributions of R_n exhibit smaller variability.

The three seasons analysed differ also in terms of magnitude, which is measured in term of Total Attack Rate (TAR), the probability of being infected over the whole pandemic for a susceptible individual. The median (95% CrI) TAR during season 2012/13 was 0.306 (0.270-0.338), which was higher in season 2013/14, 0.338 (0.313-0.363) and decreased again during season 2014/15, 0.248 (0.176-0.302).

As there was high circulation of the H3 strain (more severe, especially for the elderly [24]) in the 2014/15 season we would expect the TAR to be higher than the previous seasons, but, due to the high variability of the estimate, this is not the case.

	season 2012/13 Me(CrI)	season 2013/14 Me(CrI)	season 2014/15 Me(CrI)
π	0.559 (0.309 - 0.971)	0.547 (0.314 - 0.976)	0.538 (0.298 - 0.972)
$\lambda_0 (\times 10^4)$	0.119 (0.061 - 0.23)	0.033 (0.019 - 0.059)	0.345 (0.161 - 0.745)
ψ	0.19 (0.047 - 0.376)	0.203 (0.052 - 0.377)	0.19 (0.038 - 0.376)
η	3.394 (2.005 - 6.558)	1.235 (1.01 - 1.991)	19.961 (11.56 - 40.189)
$p_{ICU} (\times 10^4)$	0.846 (0.466 - 1.598)	0.766 (0.423 - 1.353)	1.864 (0.897 - 4.093)
R_0	2.141 (1.234 - 3.882)	2.243 (1.256 - 3.898)	2.145 (1.185 - 3.881)
R_n	1.198 (1.171 - 1.223)	1.226 (1.206 - 1.246)	1.154 (1.11 - 1.193)
I_0	1035.521 (339.782 - 2936.476)	276.546 (102.991 - 772.893)	3029.182 (928.951 - 9983.244)
TAR	0.306 (0.27 - 0.338)	0.338 (0.313 - 0.363)	0.248 (0.176 - 0.302)

Table 2: Posterior median and posterior 95% Credible Intervals of the parameters and of functions of them.

Given the importance of the basic reproduction number R_0 , as a summary of transmissibility, we also examined its posterior distribution in each season (see Supplementary Information, Section SI 4). Under the prior assumptions of Table 1, this quantity is poorly identified: its posterior is flat and a distinct posterior mode cannot be detected. This indicates a need for more

Parameter	Distribution
π	$\sim LogNorm(\log \mu = \log(0.401), \log \sigma = 0.2)$ [39]
p_{ICU}	$\sim LogNorm(\log \mu = \log(0.000239), \log \sigma = 1)$ [33] [SI 2]

Table 3: Informative prior distributions from previous findings.

informative prior distributions for the vector Θ .

We therefore repeated the analysis, using sero-prevalence data from the 2010/11 season [39] to formulate a prior for the susceptibility π . The use of sero-prevalence data to describe the immunity of a population could be debated, since the results may be extendable only to seasons

with similar predominant strains circulating. Here, sero-samples were taken during an H1 pre-dominant season: this sub-type was prevalent also in the 2012/13 season, but not in 2014/15. However, combining this prior with the data allows us to test how much prior information is needed to overcome the lack of identifiability of susceptibility from the data. An informative prior distribution on p_{ICU} was constructed by combining estimates of the probability of hospitalization given infection from a previous severity study [33] with estimates of the probability of ICU/HDU admission given hospitalization from the aggregate data of the USISS sentinel scheme (see Supplementary Information, Section SI 2). These informative prior distributions are reported in Table 3. All other parameters were assumed to follow the distributions defined in Table 1.

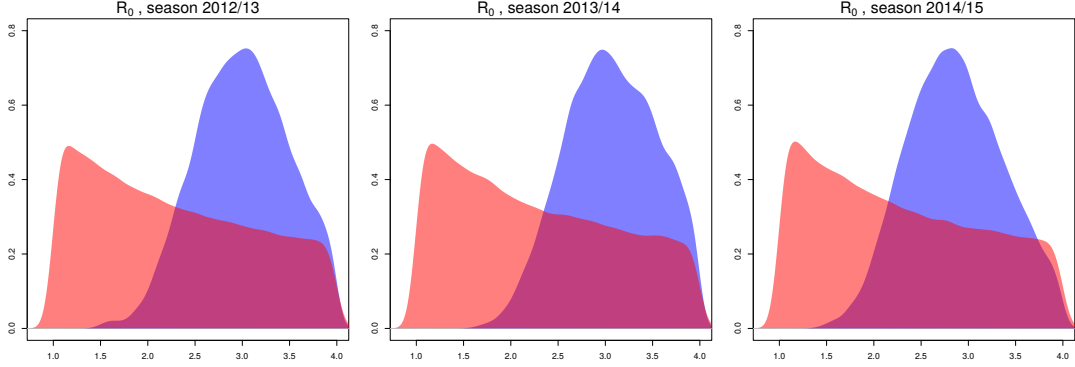


Figure 5: Prior (red) and posterior (blue) distributions of R_0 across seasons.

Posterior distributions of the basic reproduction number R_0 , based on the informative prior analysis, are displayed in Figure 5. The three posteriors look identical to each other, because they do not learn from data and are uniquely determined by the prior distributions. However, note that the posteriors differ from the priors, as R_0 is a function of ψ , which is highly correlated with π . Therefore the posterior distribution of R_0 is dominated by the prior specification of π , and hence differs from the prior distribution of R_0 (for further results with different prior specifications see Supplementary Information, Section SI 4).

4.2 Analysis of the epidemics in real time

To assess the predictive ability of our model, we performed estimation and forecasting assuming only an initial portion of the data are available. First, we used the data up to week w as a training dataset to estimate the parameters; then we predicted the evolution of the epidemic after week w , based on the estimates from the training dataset. USISS collection starts at week 40 of every year, we tested the following prediction time points: $w = 3, 8, 13$, and 18 from the beginning of the new year (which roughly corresponds to week 15, 20, 25, and 30 from the beginning of the data collection). We used the informative priors reported in Table 3.

The performance of the model at different times is plotted in Figure 6 for each season. The black line displays the analysis time; the blue line and green shaded area represent median, quartile (dark green) and 95% credible intervals (light green) of the posterior predictive distribution for the training dataset weeks. The pink area displays posterior quartiles (deep pink) and 95% credible intervals (light pink) for the predicted future observations, and the purple line displays the median; the red dots are the training data and the yellow dots are the observations we have predicted.

Season 2013/14 is the most difficult to predict: the well-defined initial growth biases the predictions towards a major outbreak. This leads to overestimation of the median and the credible intervals of the posterior predictive distribution until mid-march (week 13 from the beginning of the year). For the other two seasons, the median predicted weekly ICU admissions is always very close to the data points, but the credible intervals narrow to reasonable bounds only after the peak of each season (week 13 from the beginning of the year).

Similarly to the estimates for R_0 , the predictions are highly affected by the prior distributions on π (see Supplementary Information, Section SI 4.3). Lastly, the model used gives an idea of the size of the epidemic but it is not flexible enough to describe the specific features of each season (e.g. the double peak of season 2014/15 and the plateau of season 2012/13).

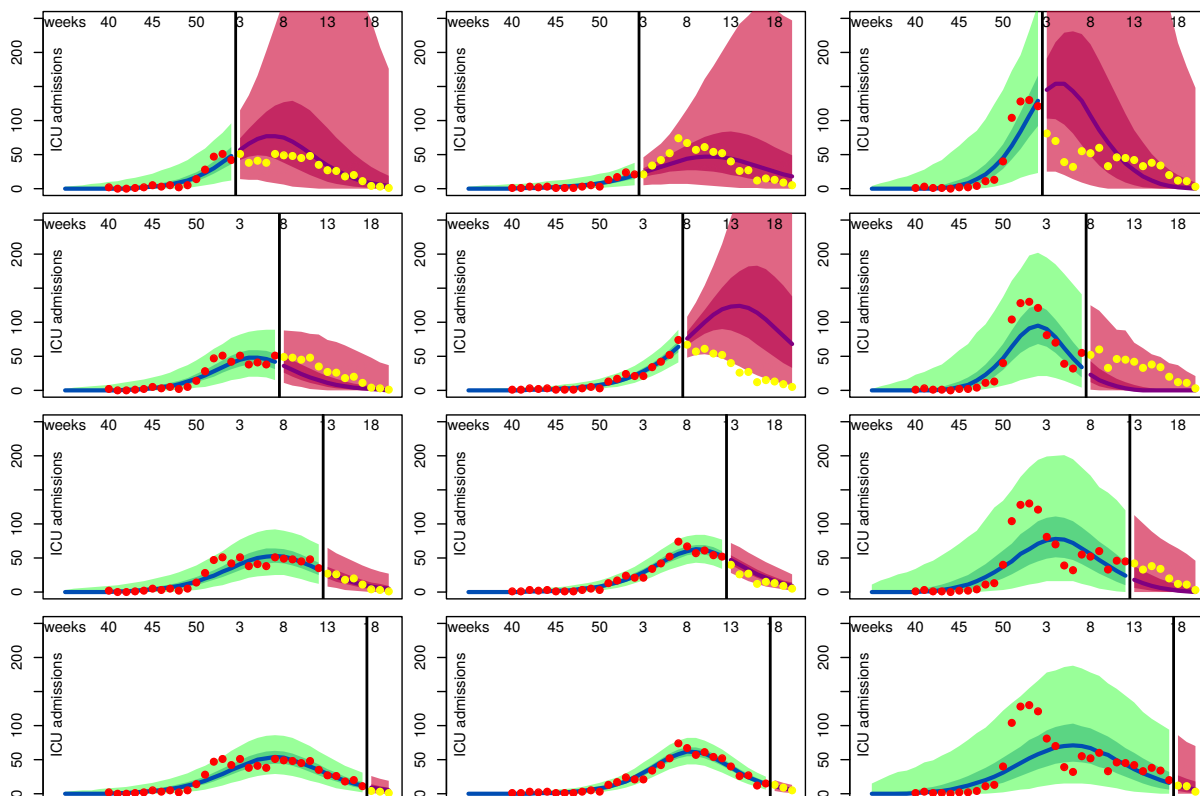


Figure 6: Evolution of the predictions of influenza: season 2012/13 (left panel), season 2013/14 (central panel), season 2014/15 (right panel).

4.3 Simulation of the analysis in the event of a pandemic

In the case of a pandemic, the weekly count of the hospitalizations for influenza at all levels of care would be available [20]. Therefore we can simplify the observational model described in Figure 2 by reducing the time from infection to entrance into the dataset, at hospital admission rather than ICU/HDU admission. Both the transmission model and the estimation methods adopted are almost identical to the ones described in Sections 3.1 and 3.2. Let p_H denote the probability of being hospitalized given infection; let $\mu_{H|E}$ and $\sigma_{H|E}$ denote the mean and the

standard deviation of the time from infection to hospitalization. These new parameters replace the former p_{ICU} , $\mu_{ICU|E}$ and $\sigma_{ICU|E}$ in the definition of the observational process.

To simulate the data, we used the transmission dynamics described in Equation 2 combined with the observational model summarized above. The set of parameters used to simulate the weekly hospitalization is the posterior median of Θ obtained from the analysis of the 2014/15 data reported in Table 2.

Then we analysed the simulated data. First we assumed the flat priors from Table 1. Second, we incorporated information defining prior distributions on parameters π and p_H . The prior distributions used were informative distributions centred on the values used for simulation. As a third analysis, we assumed data would be available in portions as explained in Section 4.2.

Despite USISS collecting more information during a pandemic, the posterior distribution of the parameters suffers from the same identifiability problem demonstrated in the analysis of seasonal epidemics, under flat priors (results shown in the Supplementary Information, Section SI 5).

This finding highlights that the weak identifiability is not due to a general lack of data, which would be improved in a pandemic situation, since data on all hospitalizations would be available. For all the parameters to be identified separately, more sources of information are necessary on the different parts of the model (e.g. one on the transmission dynamics and another on the initial immunity).

5 Discussion

In this paper we proposed a model to estimate and predict influenza outbreaks from data on admissions to ICU/HDU and, in the case of a pandemic, on all hospitalizations.

We investigated the performance of the proposed model both on simulated and on real data. By fitting the model to simulated number of weekly ICU admissions, we discovered that, even with very vague prior information, we could obtain estimates of some of the main parameters, including the initial infection rate (λ_0), the probability of going to ICU given infection (p_{ICU}), the effective reproduction number R_n and the total attack rate (TAR). When we injected information on the distribution of the average immunity ($1 - \pi$) and on p_{ICU} , estimates of the remaining parameters (e.g. the basic reproduction number R_0) could be obtained. We were also able to forecast the evolution of the outbreak by analysing just the first weeks of the epidemic using data up to the peak of influenza activity.

The model was applied to real data on weekly number of ICU admissions from seasons 2012/13, 2013/14 and 2014/15 and it confirmed the performance obtained on simulated data. The estimated values of the effective reproduction number R_n and of the TAR were similar to those estimated during the past decade of seasonal influenza [5]. However, the model was not flexible enough to reproduce some of the features in the data, such as the double peak of season 2014/15, leading to over-dispersed estimates.

Recently, a similar analysis was performed on the Finnish pandemic of 2009 [18] using a more elaborate model, including the burden of the epidemic on GPs and a time-varying transmission rate. The inclusion of GP data, in addition to ICU admissions, enhances the performance of the inference in [18]. Nevertheless, these data are harder to interpret in a pandemic setting, where health care seeking behaviour might change over time and do not reflect the underlying true signal. By contrast, the inference performed through our model is driven by data readily available, even in real time. Moreover, our analysis allows prediction of the outbreak, whereas, using the model of [18], prediction is not possible. In fact, in [18] the transmission parameter is time varying: this allows an accurate description of the past dynamics but makes the prediction

impossible since this temporal variation cannot be forecast.

Likewise, the work in [8, 13, 15] exploits data that are harder to access and describes only a pandemic setting, without considering seasonal outbreaks. Moreover, they all include in the analysis non-confirmed cases, adding a further source of noise. Similarly, [16] inferred transmission parameters combining data on health-related online queries and social media in combination with viral isolation data. These data, although being easily accessible and copious, are even more affected by social and health background noise.

Our work has also some limitations: firstly, our model is non-age-specific. This was dictated by the very small population size which did not allow sub-grouping. Secondly, the quality of some estimates and predictions strongly relies on prior information on the proportion of non-immune people. As this information is needed to overcome the lack of identifiability in the parameters, we used sero-prevalence data following the 2010/11 epidemic. This is not likely to be correct for all the three seasons analysed, as the predominant strain circulating was different across seasons. Similarly, a significant part of the observational model was fixed over different seasons, an assumption that should probably be relaxed.

Information on the immunity of the population is needed in order to obtain precise estimation and accurate early predictions. The work presented here is a proof of concept of the potential for estimation and prediction of influenza transmission from USISS data. However, at the same time, it highlights the need of the collection of external data to formulate appropriate prior on the initial immunity of the population, particularly in the event of a pandemic.

We have provided a useful tool to retrospectively infer the epidemic parameters from routinely collected data on severe cases during seasonal outbreak and to predict the temporal dynamics of new pandemics. Our model could be extended in a number of ways. Some of these are: to take into account seasonality by linking consecutive seasons and to relax constraints such as deterministic transmission dynamics or constant transmission rate.

Competing interests

We have no competing interest.

Authors' contributions

AC wrote the paper, with help by XSZ DDA and AMP. XSZ and AC wrote the code, with support by PJB and AMP in the formulation of the algorithm. NB provided the data and assisted with their interpretation. DDA and RGP conceived the study. All authors gave a final approval for publication.

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